

Lecture 29

4/3/26
JRA

Projection Radiograph

Generating "shadows" that reveal how much X-ray absorbing material intercepts the beam.

Useful but lose depth information

Can get structural superposition: lose contrast

Doing radiography - oldest technique
- CT started in 1980s

Generate X-rays or shadows

Bones - lots of absorption

ST - less absorption

Lose depth information

Get structural superposition

Lose contrast

1	1	1
1	2	1
1	1	1

3
9
3

Object has 7:1 contrast
Projection has 3:1 contrast

Doctors typically take multiple views

SIMULATION

Essence of Radiography

Looked at projections w/ cube, sphere, cylinder
3 views

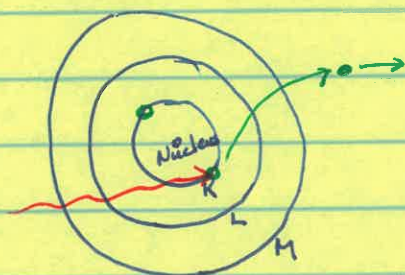
Impt to understand how ^{diagnostic} X-rays interact w/ matter

Interactions between Diagnostic X-ray & Matter

Contrast $\rightarrow$ (1) Absorption (good guy) by the photoelectric ^{effect} (PE)

(2) Scattering (bad guy) by the Compton Effect

Non-interaction (critical) is transmission thru body. This is what creates image, because need X-ray to hit detector.

PE Effect

K = most tightly bound e^-

L, M = less tightly (more weakly) bound e^-

X-ray can kick K-shell electron out of element
Ca, H, O

X-ray disappears
Electron kicked out w/ KE
dependent on X-ray energy

This gives contrast because beam intensity is reduced

Key Point

Random - probability that x-ray will interact w/ atom

Couldn't find derivation

Probability $\propto \rho \frac{Z^3}{E^3}$ density

ρ = density
 Z = atomic # of atom or element
 E = energy of x-rays

Want a lot of PE to get image

Not 100% because need

$\Rightarrow \begin{cases} Z \text{ big} \\ E \text{ small} \end{cases}$ to improve contrast

Example of how to get good contrast

When look at radiograph, it's easy to tell bone vs soft tissue

bone has a lot of Ca^{2+} $Z = 20$

bone $Z_{\text{effective}} \approx 13$

$Z_{\text{effective}} \approx 6$
 ST

Same E

$$\frac{\text{Prob|}_{\text{bone}}}{\text{Prob|}_{\text{ST}}} = \frac{P_{\text{bone}} \frac{Z_{\text{eff bone}}^3}{E^3}}{P_{\text{ST}} \frac{Z_{\text{eff ST}}^3}{E^3}} \quad \frac{P_{\text{bone}}}{P_{\text{ST}}} = \frac{P_{\text{bone}}}{P_{\text{ST}}} \left(\frac{13^3}{6^3} \right) \approx 2^3$$

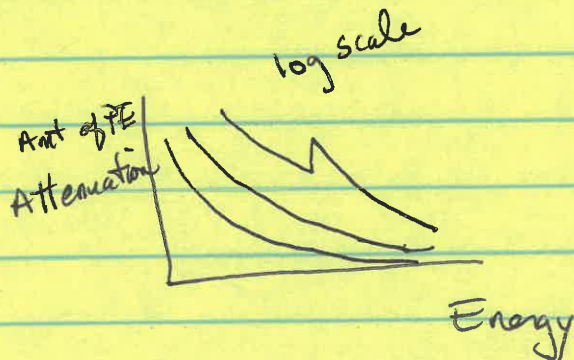
$$P_{\text{bone}} > P_{\text{ST}}$$

Much higher Z_{eff} for bone causes it to absorb much more

What about energy effect?

Dr can manipulate the energy of the beam

Good contrast comes from lower E



Difference between bone & ST decreases as energy increases
Attenuation becomes more similar

Wants to maximize contrast doctor will go w/ low E

To get X-rays thru body need high E

Trade off low E is less penetrating
Need something

Can't go too low, especially if thick
Need some transmission to get image

Example 1

Mammogram - compress breast to make it thinner

Example 2

Contrast agents - in many cases in X-ray imaging use a high Z element to add ^{selective} contrast

Blood - iodine

GI tract - barium

These increase PE effect. Creates contrast between blood $\hat{e}$, surrounding tissue.

Compton scattering is bad guys.

Drs use anti-scatter grid to reduce detected scatter

X-rays are assumed to be following straight-line trajectories

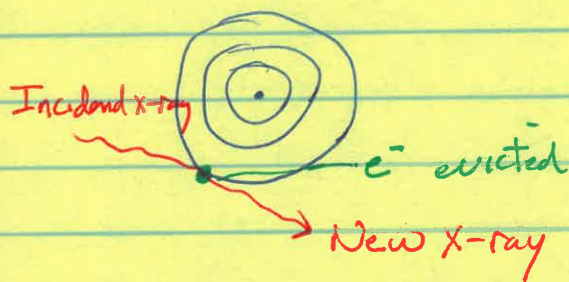
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Want correlation between position on image
& trajectory

Scatters X-rays in crazy direction
Creates random background 'fog' that reduces contrast

PE effect - X-rays kicks out K-shell e^-
Disappears

Compton



X-ray knocks out
outer shell electron
Doesn't lose a lot
of energy
X-ray scatters
(thrown off trajectory)

This can be a dominant effect

Want big PE relative to Compton
Occurs when go down w/ E

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Images



Blur shadow

Fixes

Use anti-scatter grid



Grid

Minimize ^{detection} w/ anti-scatter grid

Local Contrast

Doctors define a quantity characterizing ability to distinguish object that is absorbing relative to background

Is there a disparity between object & background

$$C_{\text{local}} = \frac{I_b - I_o}{I_b}$$

I_b ← Normalize to make it dimensionless

Intensity falls off exponentially w/ distance x in body: Beer's Law

$$I(x) = I_o e^{-\mu x}$$

attenuation coeff

Activity 10

Local Contrast

Background = ST

(A) $C_{\text{local}} = 0.02$ NO contrast Contrast NOT effective because wrong energy

(B) $C_{\text{local}} = 0.15$ Contrast agents effective

(C) Yes 7x higher contrast

$$C = 1 - e^{-(0.6-0.4)(0.1)} = 1 - 0.98 = 0.02$$

$$1 - 0.85 = 0.15$$

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